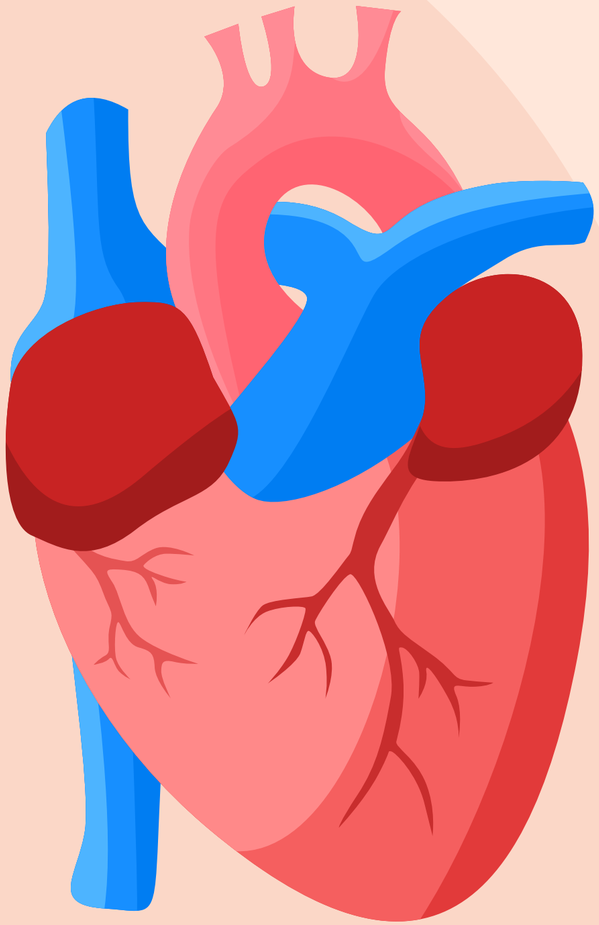
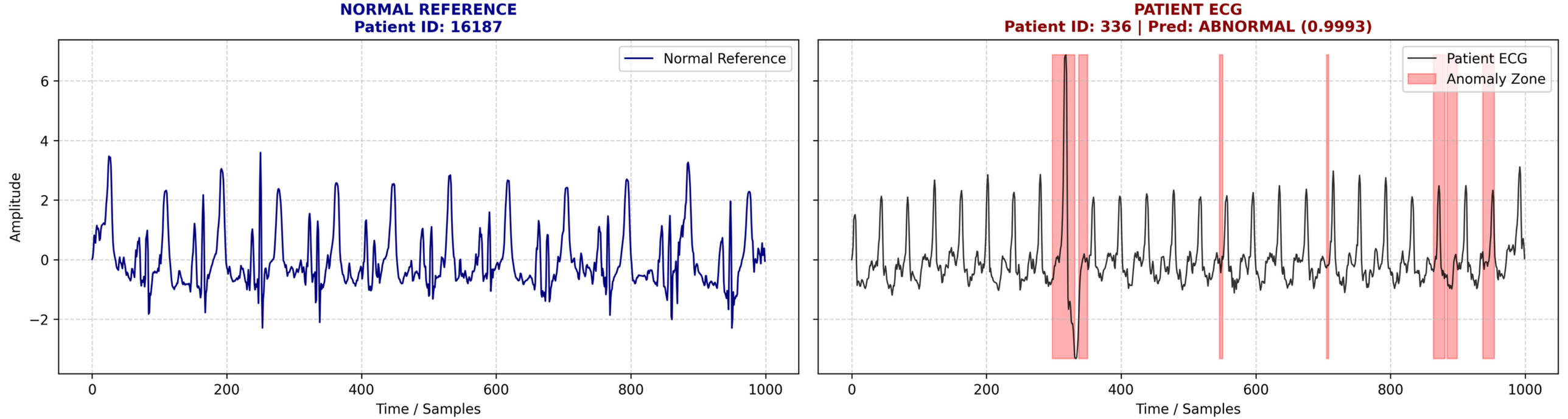


ECG ANOMALY DETECTION

Group 15

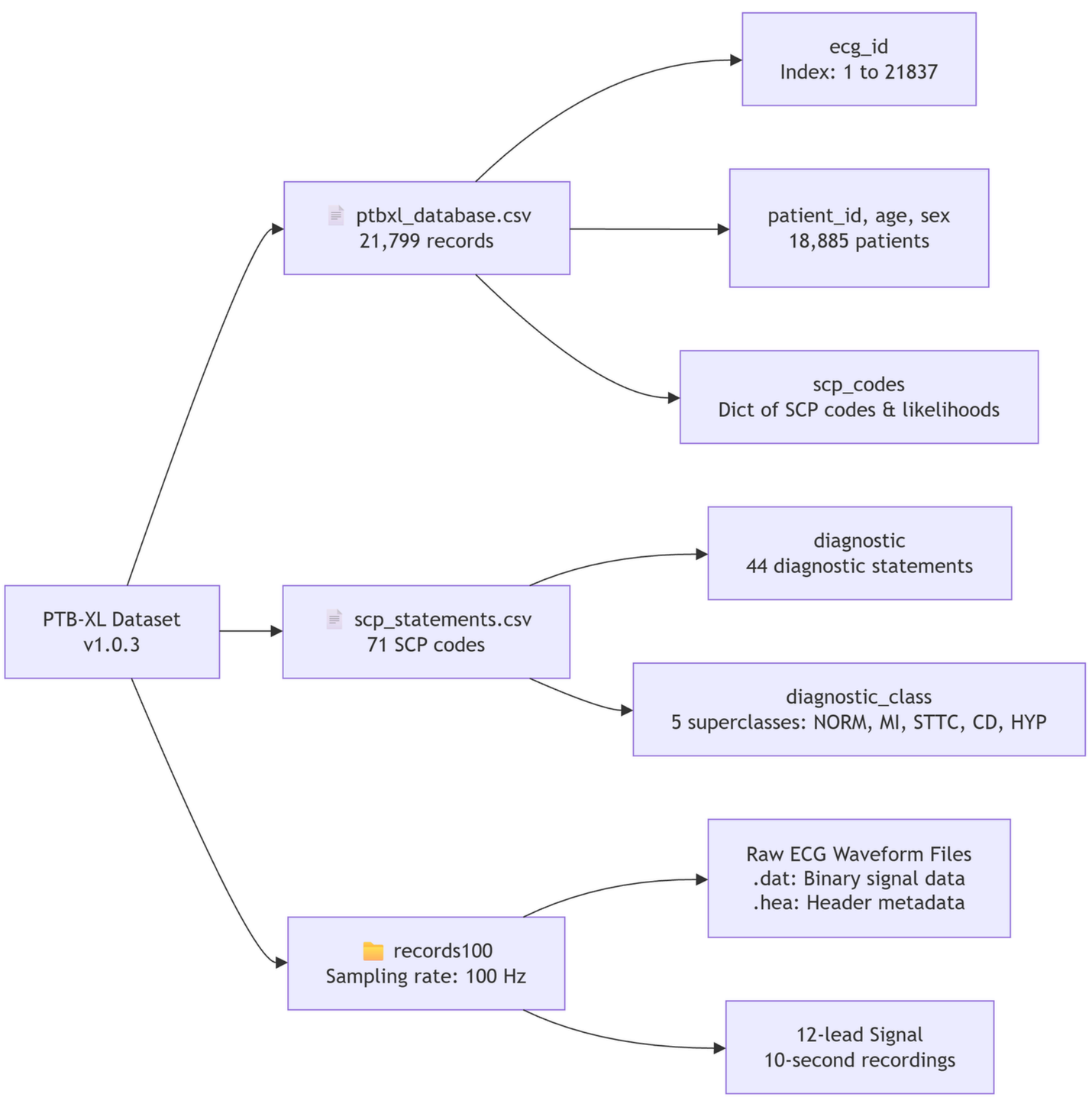
● Problem Statement



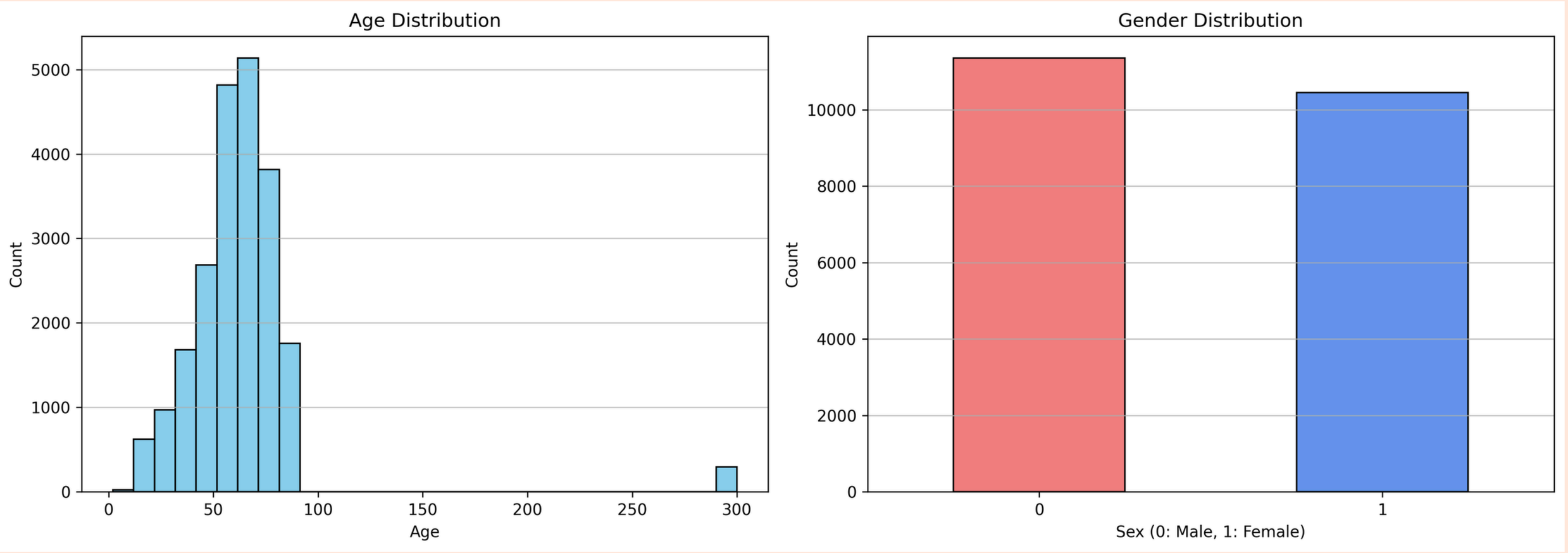
● Dataset

Physikalisch-Technische Bundesanstalt
Braunschweig und Berlin

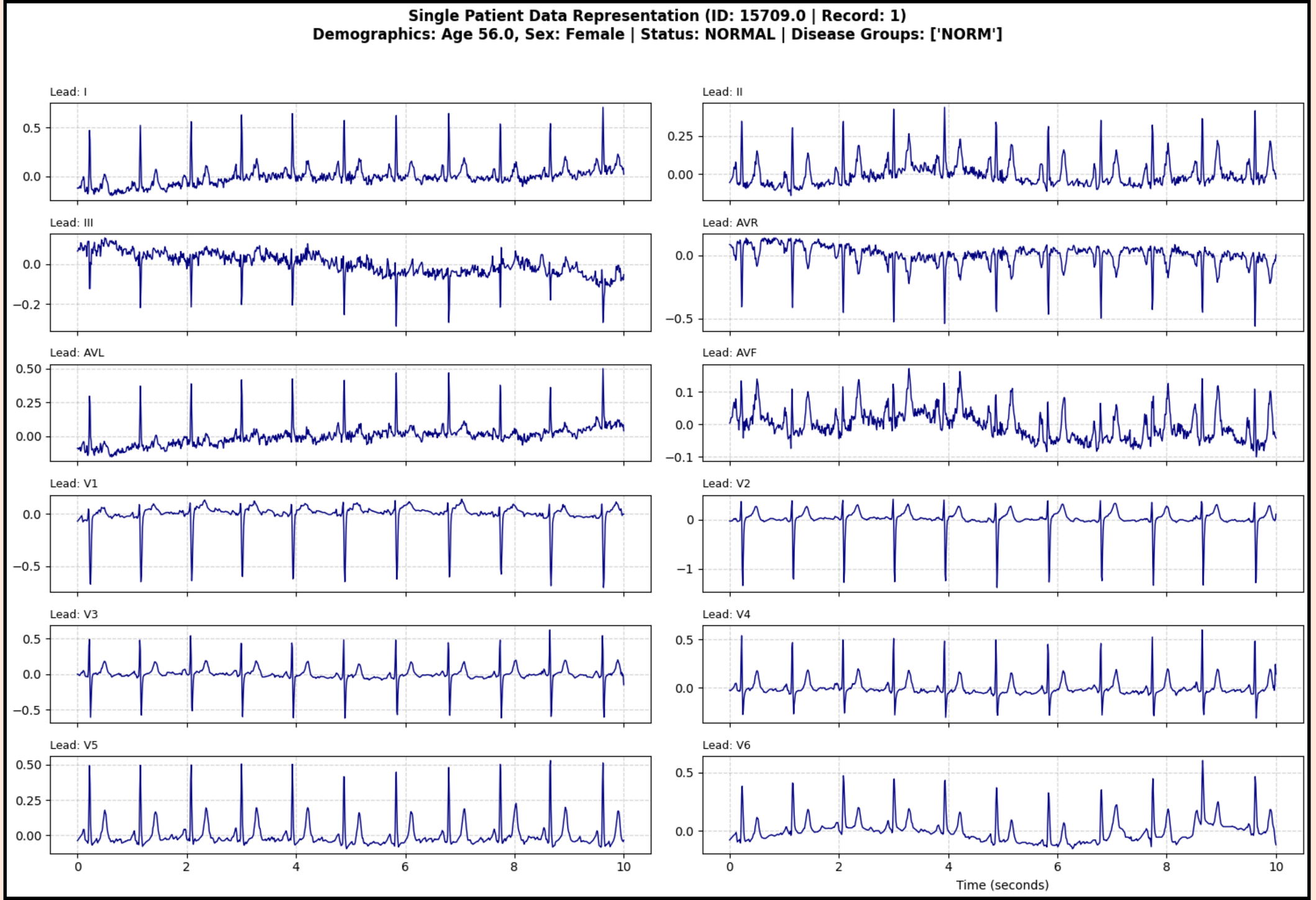
Authors: Patrick Wagner,
Nils Strodthoff.
10/1989 - 06/1996



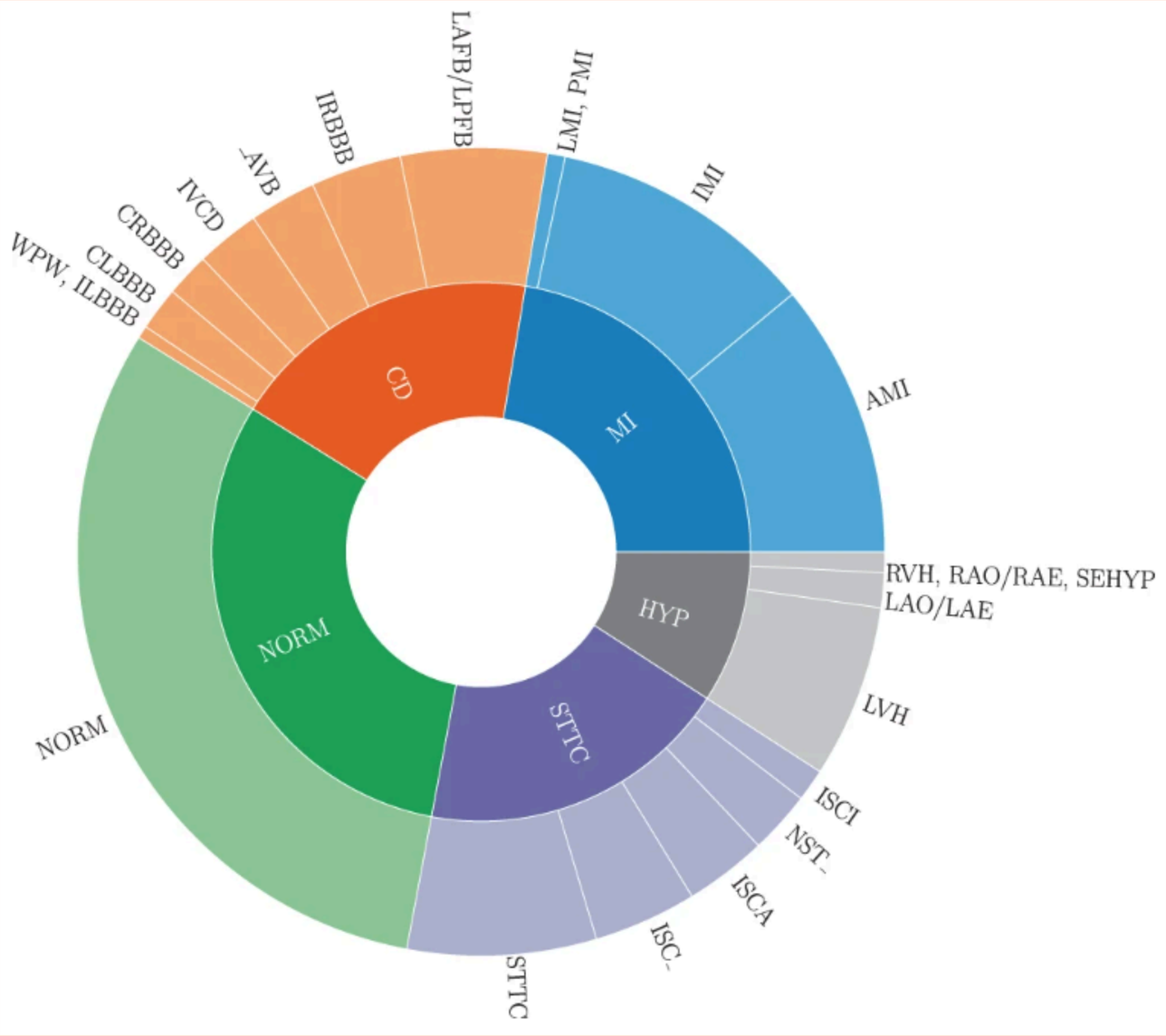
● Data Properties



Patient Demographic Distribution


Single Patient Data Representation (ID: 15709.0 | Record: 1)
Demographics: Age 56.0, Sex: Female | Status: NORMAL | Disease Groups: ['NORM']
Lead: I
Lead: II
Lead: III
Lead: AVR
Lead: AVL
Lead: AVF
Lead: V1
Lead: V2
Lead: V3
Lead: V4
Lead: V5
Lead: V6
Time (seconds)

Ontology & Disease Group Mapping



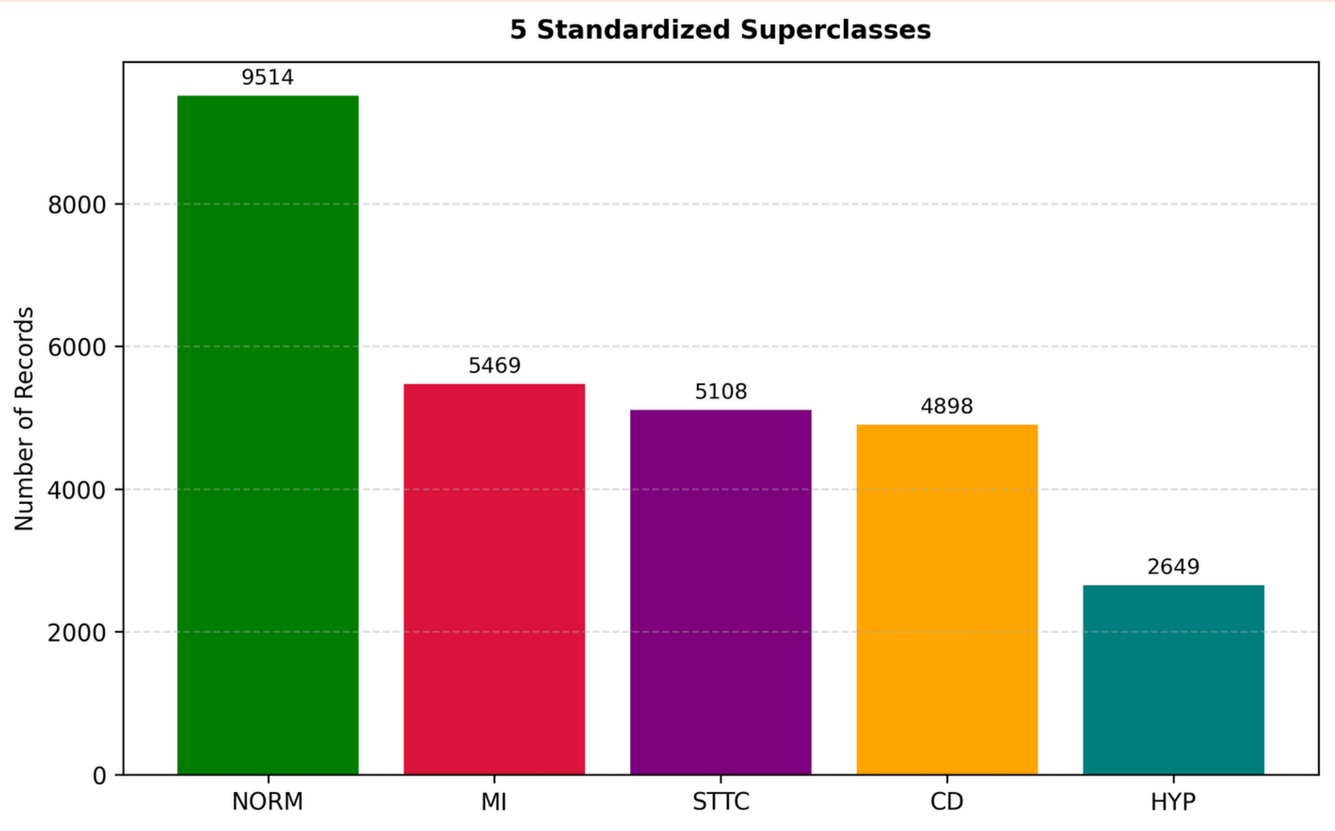
Ontology & Disease Group Mapping

ptbxl_database.csv

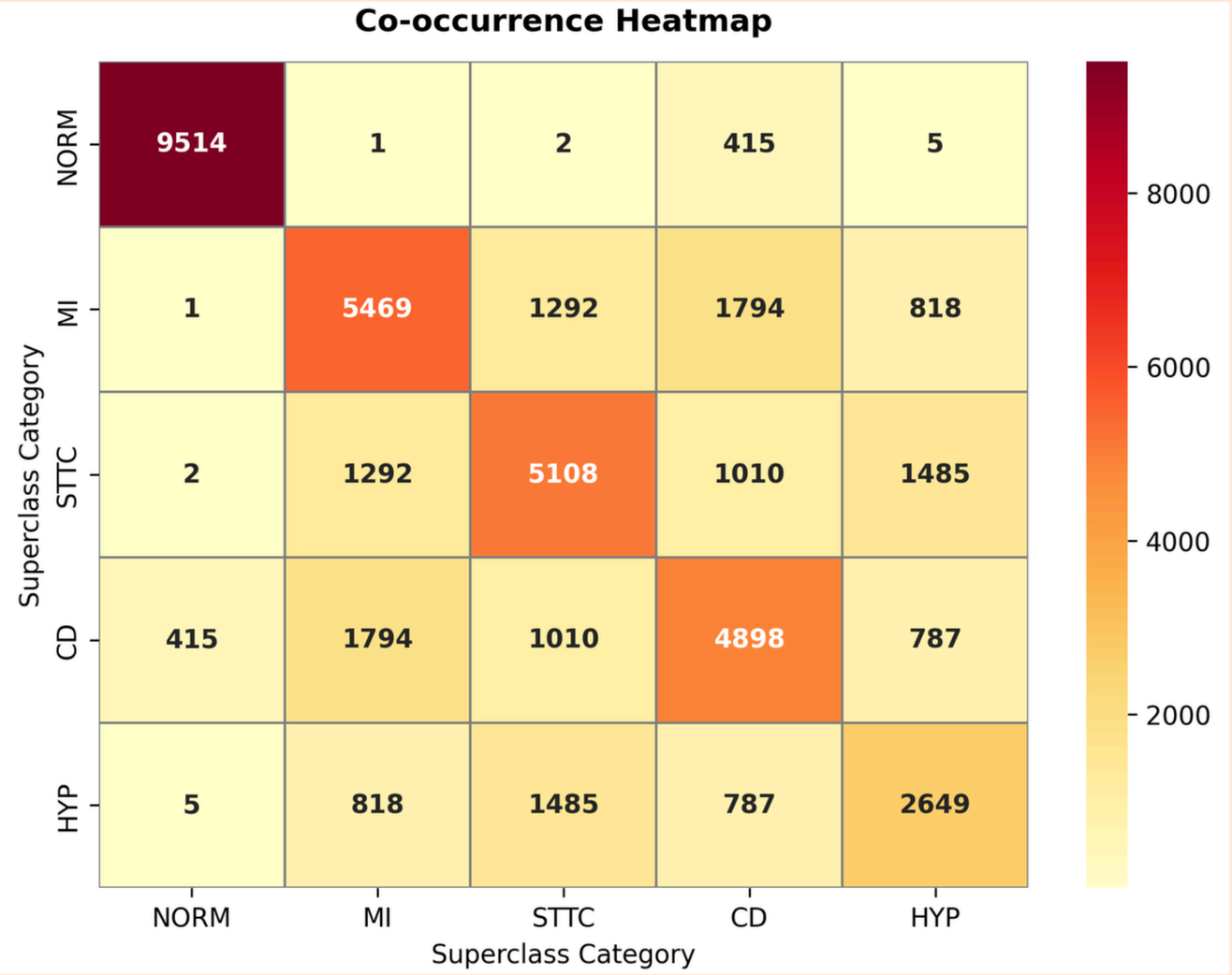
ecg_id	patient_id	age	sex	height	weight	nurse	site	device	recording_date	report	scp_codes
1	15709.0	56.0	1		63.0	2.0	0.0	CS-12 E	1984-11-09 09:17:34	sinusrhythmus periphere niederspannung	{'NORM': 100.0, 'LVOLT': 0.0, 'SR': 0.0}

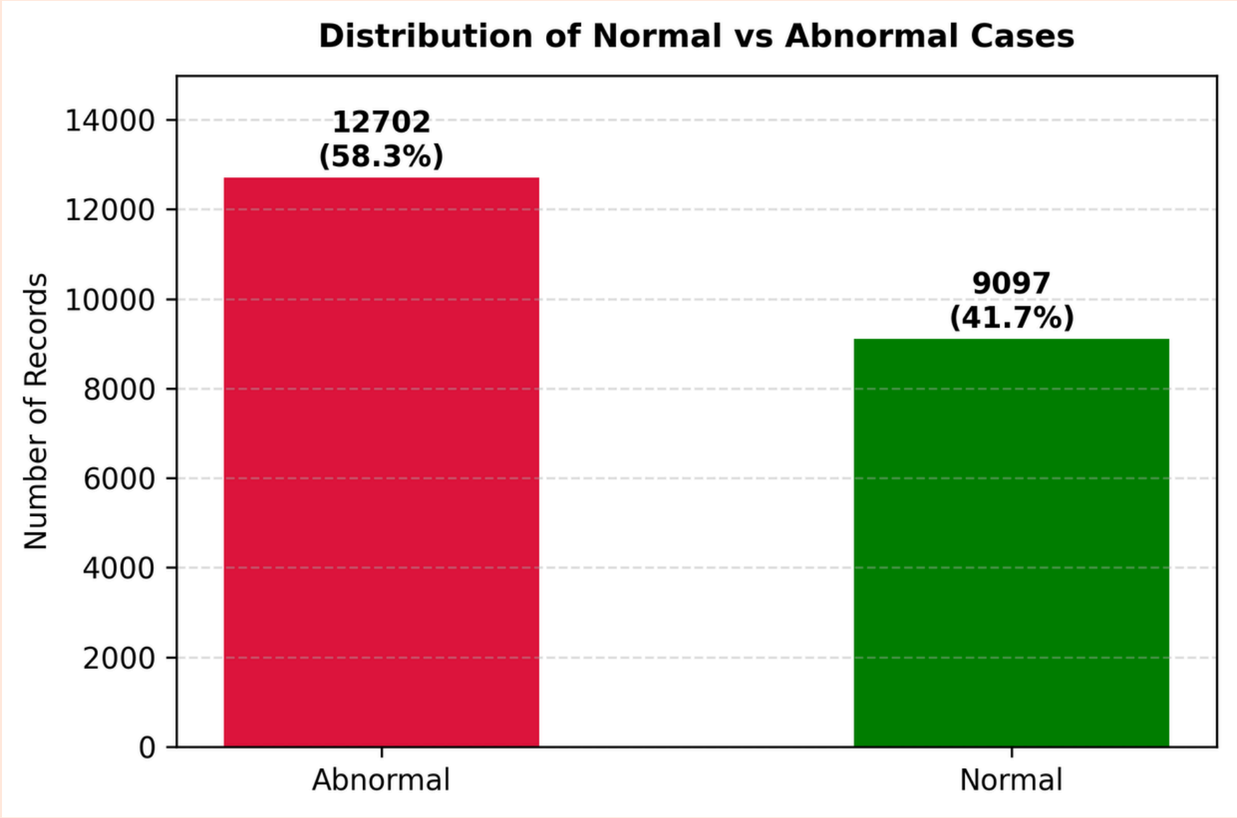
scp_statements.csv

	description	diagnostic	form	rhythm	diagnostic_class
NDT	non-diagnostic T abnormalities	1.0	1.0		STTC
NST_	non-specific ST changes	1.0	1.0		STTC
DIG	digitalis-effect	1.0	1.0		STTC
LNGQT	long QT-interval	1.0	1.0		STTC
NORM	normal ECG	1.0			NORM

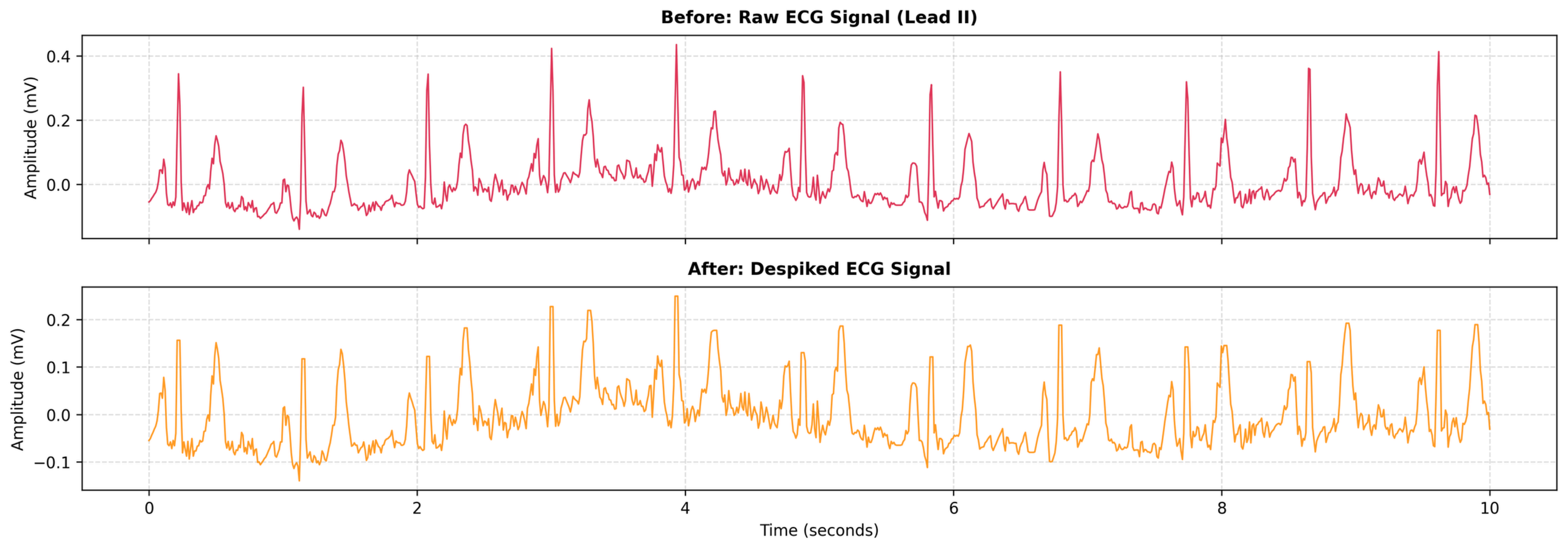


Ontology & Disease Group Mapping

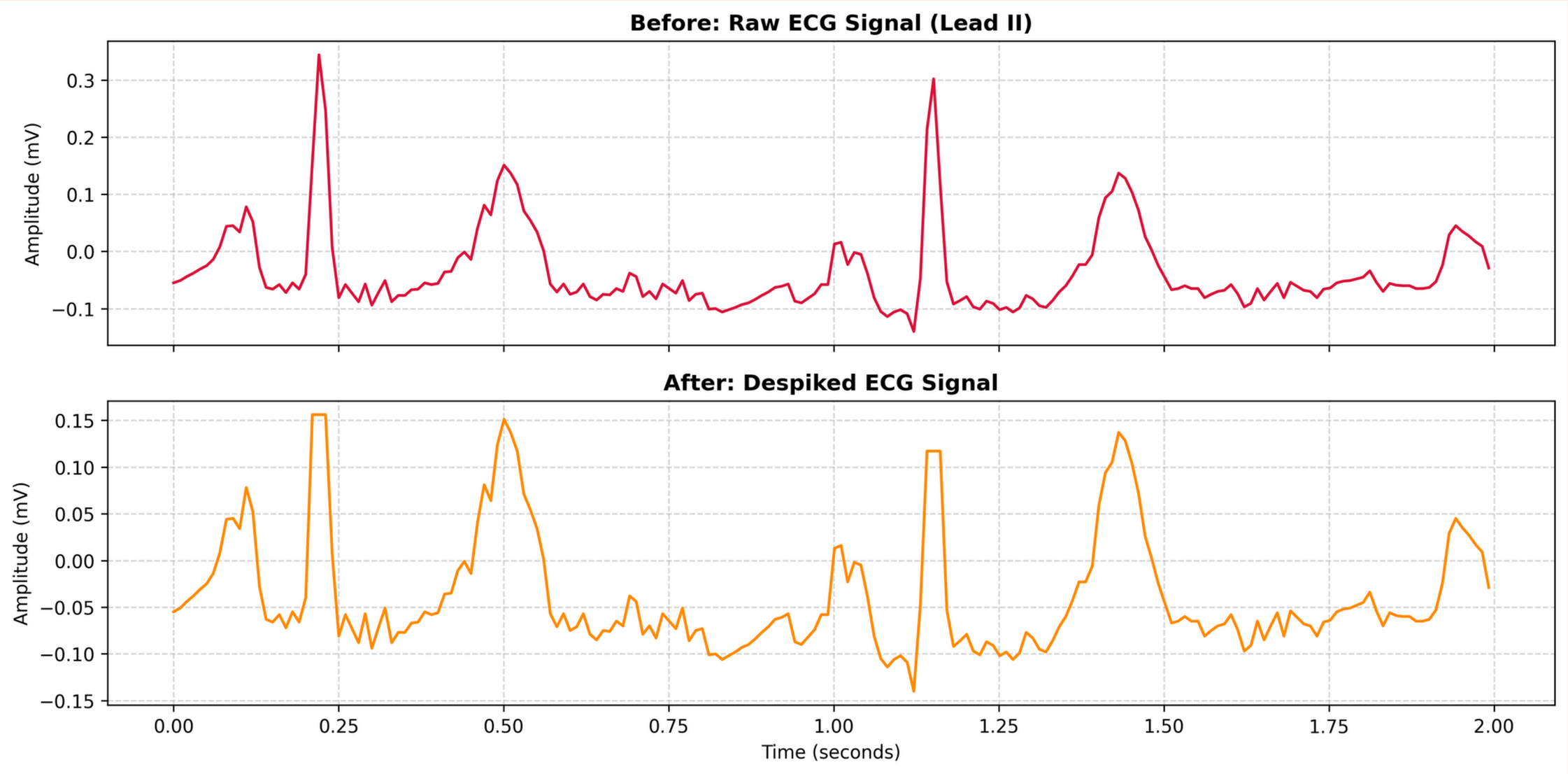




Noise Removal

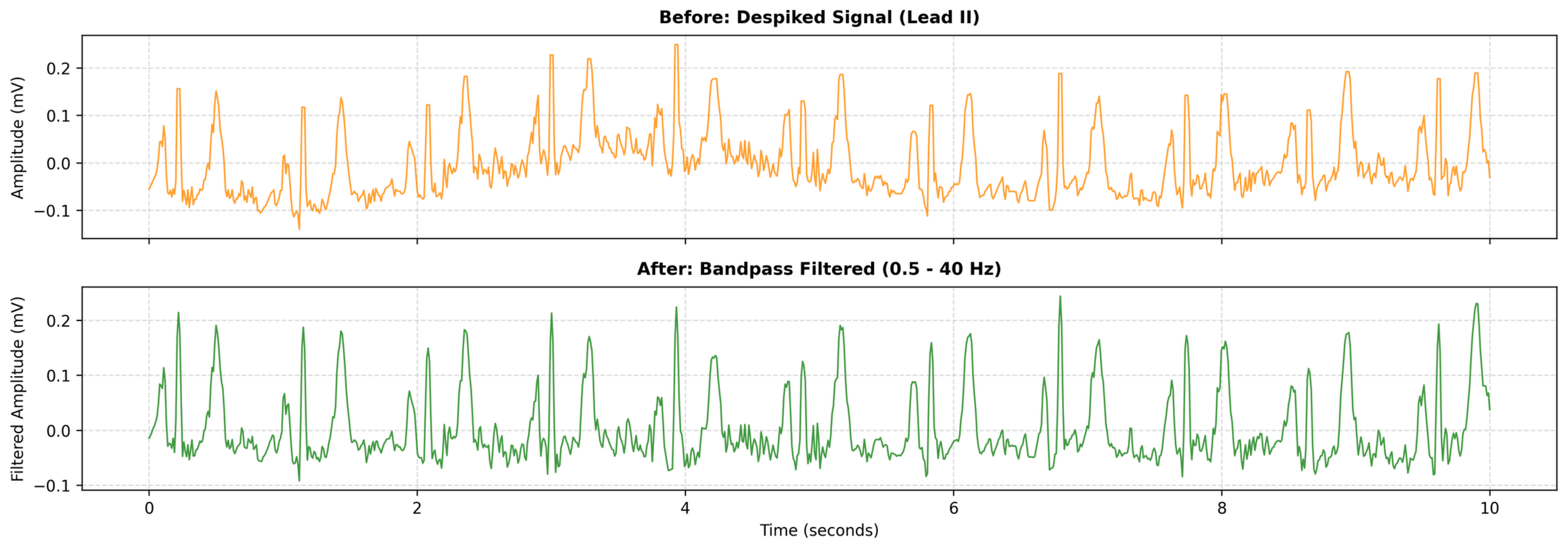


Noise Removal

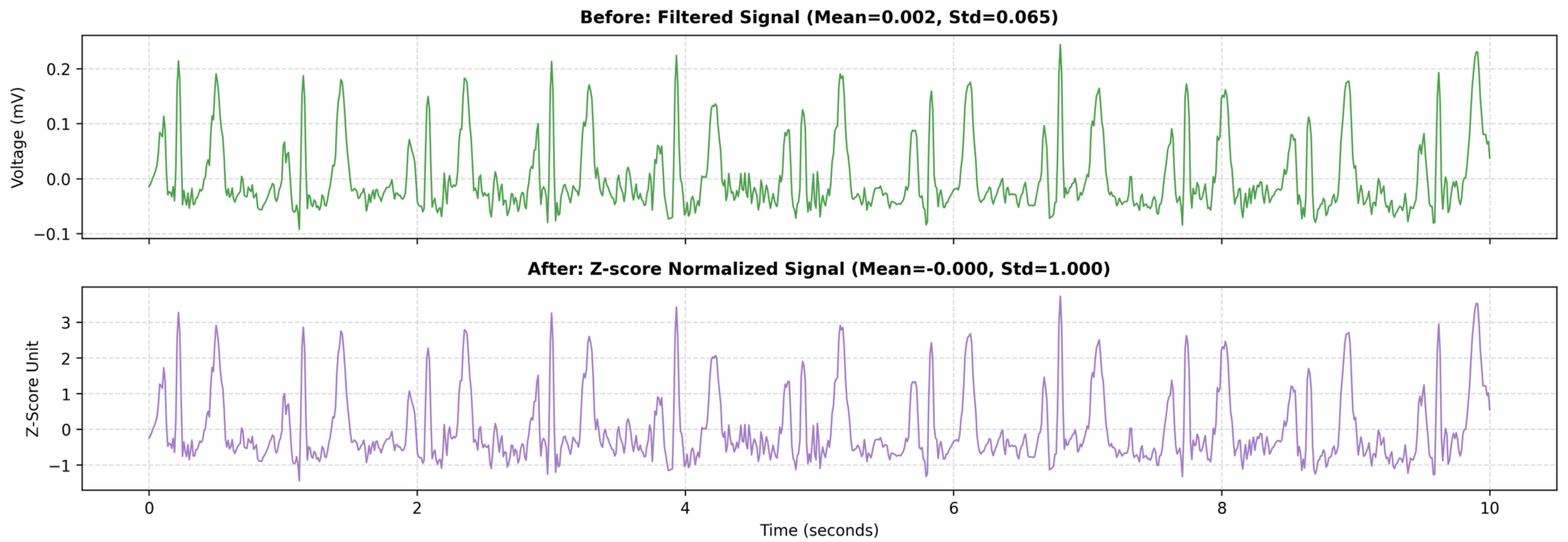


Zoomed 2 seconds

Noise Removal



Scaling


Before: Filtered Signal (Mean=0.002, Std=0.065)
Voltage (mV)
0.2
0.1
0.0
−0.1
After: Z-score Normalized Signal (Mean=-0.000, Std=1.000)
Z-Score Unit
3
2
1
0
−1
0
2
4
6
8
10
Time (seconds)

Q&A Section



THANK YOU

